

CLAUDIO MALVINO

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EDUCATION

University of Cambridge

Oct 2025 – Sep 2026

MPhil in Scientific Computing

Relevant Coursework: High Performance Computing, Numerical Methods, Scientific Computing, GPU Programming

CUNY New York City College of Technology

2021 – 2023

B.S. Applied Computational Physics, magna cum laude

GPA: 3.76/4.0

RESEARCH EXPERIENCE

MPhil Research Project – Performance Optimization of CP2K

🔄 Merged PR | Fork

University of Cambridge | Supervisors: Dr. Christoph Schran, Dr. Peter Cooke

Apr 2026 – Aug 2026

- Redesigned the OpenMP parallelization of CP2K's neural-network potential forces and merged it into the main CP2K repository (PR #5747): threaded the atomic-centre loop with per-thread workspaces and a fixed-order reduction, delivering a $42\times$ speedup over serial on 76 cores ($57\times$ over the previous threaded path) with forces bit-identical to serial at any thread count.
- Profiled the machine-learned interatomic potential (Behler–Parrinello neural network) force evaluation with MAQAO and hardware-counter analysis on the CSD3 supercomputer, isolating the dominant compute bottlenecks.
- Developed two optimized branches of the Fortran/C codebase achieving $5.6\times$ single-core and $2.6\times$ full-node (76-core) speedups on a 3,072-atom bulk-water system by replacing an $O(N^2)$ pair search with Verlet neighbour/cell lists, adding cubic Hermite spline interpolation, and tuning cache, loop, and OpenMP behavior – reproducing upstream energies and forces bit-for-bit.

REU Fellow

May 2023 – Jul 2023

University of Texas at Dallas | Supervisor: Dr. Michael Kolodrubetz

- Built Python simulations of driven 2D hexagonal lattices with tight-binding models to study non-equilibrium (Floquet) topological phases.
- Computed transport properties and identified phase transitions across Floquet, Anderson, and Haldane regimes.

Undergraduate Researcher

Jan 2023 – Sep 2023

CUNY New York City College of Technology | Supervisor: Dr. Roman Kezerashvili

- Derived closed-form solutions to the 2D Schrödinger equation for central and Kratzer-type potentials via the Nikiforov–Uvarov method, contributing to two peer-reviewed publications.
- Implemented symbolic and numerical computations and 3D probability-density visualizations in Mathematica.

PROJECTS

Monte Carlo Options Pricer (C++20, OpenMP, pybind11)

🔄 Source

- Built a Monte Carlo engine pricing European call/put options under geometric Brownian motion, exposed to Python via pybind11 with type stubs and a CMake build.
- Parallelized the simulation with OpenMP (per-thread RNG streams + reduction), achieving $\sim 4\times$ speedup on a 4-core/8-thread CPU (near-linear to the physical cores); the serial C++ engine runs $\sim 3.3\times$ faster than pure Python.
- Designed an immutable, const-correct option class with input validation and deterministic per-thread seeding for reproducible, statistically consistent prices.

Parallel Molecular Dynamics Simulation (C++20, MPI – Cambridge MPhil)

- Implemented a distributed-memory MD simulation (Lennard-Jones, velocity-Verlet) parallelized with MPI across up to 32 cores using particle decomposition and custom MPI struct datatypes.
- Ran strong-scaling studies (500–2,000 atoms) quantifying speedup and parallel efficiency; verified energy conservation to under 1% drift.

Physics TUI (Python, Textual)

🔄 Source

- Built and published an interactive terminal app exposing 150 physics equations across 12 domains, with 58 multi-variable interactive solvers.
- Engineered for rigor: 82 `pytest` unit tests, strict `mypy` typing, Black formatting, and a modular inheritance-based architecture.

Physicalc (Kotlin Multiplatform, Compose)

Source

- Cross-platform physics study app for Android and desktop featuring interactive equation solvers, a 3D vector calculator, and SM-2 spaced-repetition flashcards.
- Ported from Physics-TUI using AI-assisted translation into Kotlin, with behavior validated against the original Python implementation.

Stochastic Ground-State Eigensolver (Python, Qiskit – Cambridge MPhil)

- Implemented a Monte Carlo Projected Quantum Eigensolver (MC-PQE) using importance-sampled Hamiltonian terms and walker population control.
- Reached milli-Hartree (chemical) accuracy on H_3^+ validated against full configuration interaction, with reblocking-based statistical error analysis.
- Built a modular Qiskit/PySCF pipeline (UCCSD ansatz, Jordan–Wigner mapping) with profiling and convergence studies.

Tensile (C++23, in progress)

Source

- Building a header-only linear algebra library: expression templates for lazy, single-pass evaluation without temporaries, cache-optimized BLAS-style kernels, LU/Cholesky/QR decompositions, and 64-byte-aligned allocators for SIMD, with tests and benchmarks running under CI.

PROFESSIONAL EXPERIENCE

Logistics Coordinator2025

Reliance Aerospace Assets | Remote – Fort Lauderdale, FL

- Sourced commercial aircraft parts and brokered their resale, evaluating purchases by comparing acquisition and repair costs against market resale prices.
- Coordinated the repair-and-return cycle with certified repair stations, tracking turnaround times and maintaining full documentation and traceability.
- Managed quotes, invoicing, and shipping logistics across concurrent transactions while working fully remotely and self-directed.

Physics TutorJan 2024 – Dec 2024

CUNY New York City College of Technology | New York, NY

- Tutored undergraduates in introductory and intermediate physics, one-on-one and in small groups, adapting explanations of problem-solving methods to each student’s level.

PUBLICATIONS

- Kezerashvili, R.Y., Luo, J., Malvino, C.R. *On an Exactly Solvable Two–Body Problem in Two–Dimensional Quantum Mechanics*. Few-Body Systems (2023).
- Kezerashvili, R.Y., Luo, J., Malvino, C.R., Spiridonova, A. *Kratzer Potentials in Two Dimensions: Exact Solutions and Exciton Applications*. Int. J. Theor. Phys. (2025).

TECHNICAL SKILLS

Languages: C++ (17/20/23), Python, Fortran, Rust, Bash
Libraries: STL, OpenMP, MPI, CUDA, Eigen, Boost, pybind11, NumPy, SciPy, pandas, Qiskit
Tools: CMake, Git, GitHub Actions, GDB, Valgrind, MAQAO, Slurm, Linux, JupyterLab
Methods: Monte Carlo simulation, numerical methods, parallel & distributed computing (MPI/OpenMP), performance profiling & optimization, stochastic processes, AI-assisted development validated through tests & benchmarks

HONORS & AWARDS | NSF REU Fellowship • Daria Boudana Memorial Scholarship • Emerging Scholars Program • Honors Scholars Program